BENNETT UNIVERSITY

School of Computer Science Engineering and Technology

B.Tech Computer Science and Engineering, Semester V

SCSE3040 • Machine Learning Operations

Lecture 03

First Predictions

Two questions, one honest score

Which question are we asking? → What did we hide from it? → What is the number to beat?

three questions that decide whether a score means anything

Recap of Lecture 02 in three points. One project all semester: predict delivery time in minutes. We agreed the score before building, and lower is better. Eight tools, one job each, in a fixed order.

You now have a working environment and a problem statement. Today we make the first real prediction, and learn how to judge it honestly.

Issue 1.0, 21 August 2026 • Every source cited was verified on 21 August 2026

Every score in this handout was measured on the course dataset, not copied from a book

Contents

How this handout maps to the approved session plan 4

I Which Question Are We Asking? 5

1 Two kinds of question 5

2 The map scikit-learn gives you 5

3 So which one is our problem? 6

4 Common errors in choosing the question 6

II Train and Test 7

5 You cannot mark your own answer sheet 7

6 Hiding data from your own model 7
6.1 Splitting, in three lines . . . 7

7 What overfitting actually looks like 7
7.1 Why it happens, and what the curve looks like . . . 8

8 Leakage: the error that flatters you 9

9 Beyond a single split: cross validation 9

10 Common errors in splitting 10

III Scoring Honestly 11

11 Accuracy is not one number 11

12 Two scores, both in minutes 11
12.1 Measuring both . . . 11

13 The number to beat 12

14 If it had been classification 12

15 Common errors in scoring 13

16 What our project gains today 13

IV Consolidation 14

17 Assignment questions, as released 14

18 Check your understanding 14

19 Vocabulary introduced today 15

V The Week Ahead 16

20 Seven days, about seven hours 16

VI Resource Library 18

21 For this lecture specifically 18

22 Carried forward 18

List of Figures

1 The first decision of any supervised project. Regression predicts a number on a scale; classification predicts which group something belongs to. 5
2 The first fork of the scikit-learn estimator map. Everything in this course sits on the right hand branch, because our answer is a number of minutes. 6
3 Our 600 rows, split. The proportions matter less than the discipline: the model must be scored on rows it has never met. 7
4 Overfitting, measured on our own dataset. The healthy sign is not a low training score; it is two numbers that agree with each other. 8
5 The shape every practitioner carries in their head. You cannot see the dashed curve without a test set, which is the entire argument for hiding data from yourself. 8
6 The same four operations in two orders. Only the sequence differs, and only one of the two produces a number you can put in a report. 9
7 Five fold cross validation. Every row is used for training four times and for scoring exactly once, which is why the estimate is steadier than a single split. 10
8 The gap between the two scores is itself a diagnostic. Reporting only one of them throws that diagnostic away. 11
9 The baseline is what converts a score into evidence. An MAE of 1.92 means nothing until the reader knows that doing nothing scores 10.32 on the same 120 orders. 12
10 The confusion matrix, and the three scores read from it. Regression has no equivalent, which is why accuracy cannot be quoted for our project. 12
11 Still a notebook, but now a notebook with evidence in it. Nothing later in the course is allowed to break the honesty of that number. 13

How this handout maps to the approved session plan

Topic in this handout	Session plan location	Note
Train and test methodology	Session 5	The core of the session, extended here with leakage and cross validation
Evaluation metrics for regression and classification	Session 5	Regression in depth; classification metrics previewed, since Lecture 04 uses them
Activity: select and justify the right metric for delivery time prediction	Session 5, the named activity	Sections 12 to 14, and the assignment set in Part IV
Model families	Session 5	*Not here.* The released deck places the family of models in Lecture 04, and this handout follows the released material
Model artifacts and baselines	Session 6	Baselines are introduced here because a score is meaningless without one; artifacts follow in Lecture 04

The taught hour, minute by minute

0 to 4	Recap in three points, then Section 1, the two kinds of question
4 to 14	Sections 2 to 4, naming them, the map, and why ours is regression
14 to 22	Sections 6 and 7, the exam analogy and the split. Write the 480 and 120 on the board
22 to 34	Sections 9 and 10, overfitting with the measured numbers, then leakage. Do not rush leakage
34 to 42	Sections 12 and 13, MAE against RMSE, and what the gap tells you
42 to 52	Section 14, the number to beat. This is the centre of the lecture
52 to 60	Section 15, the classification preview, and the summary

Section 11, cross validation, is optional in class and assumed in the week programme.

PART I

Which Question Are We Asking?

1 Two kinds of question

Almost every supervised model answers one of two questions.

REGRESSION

"How much?"
A number.
How many minutes will this order take?
27.4, 31.8, 44.2

CLASSIFICATION

"Which one?"
A label.
Will this order be late: yes or no?
late, on time

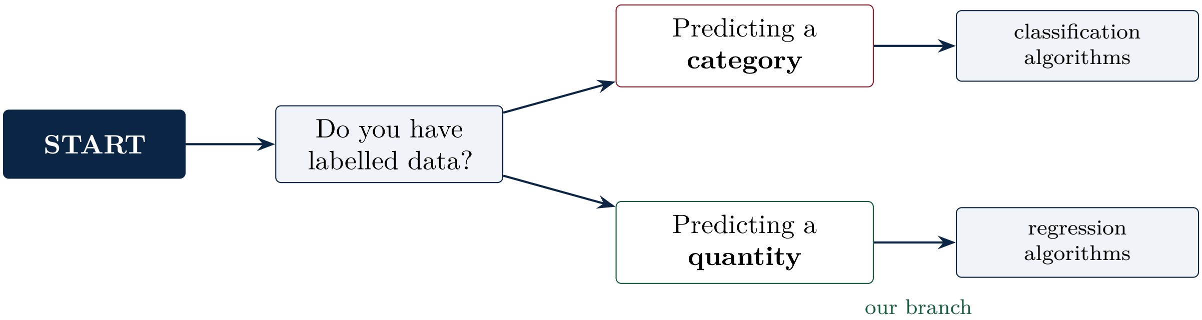
Same data, same customer, a different question, and a completely different model. The choice is made by you, before any code, and it rules out half the algorithms and every score belonging to the other side.

Figure 1: The first decision of any supervised project. Regression predicts a number on a scale; classification predicts which group something belongs to.

The task	Which kind	Why
A student's marks out of 100	Regression	A number on a scale
Pass or fail	Classification	Two groups
Tomorrow's temperature	Regression	A number, and the scale is meaningful
Marking an email as spam	Classification	Two labels
How many orders a restaurant gets on Sunday	Regression	A count, still a number
Whether a delivery will be late	Classification	Two labels, derived from a number

2 The map scikit-learn gives you

The library ships a decision map for exactly this choice. Follow START, then labelled data, then the branch: predicting a category on one side, predicting a quantity on the other.



Source: scikit-learn, Choosing the right estimator. The map continues far beyond this point, but the first fork is the one that decides everything downstream, including which scores you are allowed to quote.

Figure 2: The first fork of the scikit-learn estimator map. Everything in this course sits on the right hand branch, because our answer is a number of minutes.

3 So which one is our problem?

Our answer column is `delivery_min`, and it holds values such as 27.4, 31.8 and 44.2. Those are numbers on a scale, not groups.

The decision, recorded

The delivery time predictor is a **regression** problem.

That single decision rules out half the algorithms in the map, and every classification score you have ever heard of. Accuracy, precision and recall do not apply to us, and quoting them would be a category error rather than a small mistake.

4 Common errors in choosing the question

The error	Why it costs you
Turning a number into groups for no reason	Fast, medium and slow throws away information you already had. Thirty one minutes and thirty nine minutes become the same answer
Reporting accuracy for a regression model	There is no correct answer to be accurate about. A prediction of 31.9 is not wrong when the truth was 32.0
Assuming more classes is harder	Sometimes the right question really is a label. Ask what the user needs to decide
Choosing the question to suit the algorithm you like	The decision belongs to the user of the output, not to the modeller

When converting a number into a label is legitimate

There is one good reason to turn regression into classification: when the *decision* downstream is binary and a threshold already exists in the business. If the application shows a warning banner only when an order is predicted to exceed forty minutes, then a late or not late model may be the honest formulation, because that is the decision being made. Convert because a decision demands it, never because groups feel easier to score.

PART II

Train and Test

5 You cannot mark your own answer sheet

Imagine a student given the question paper a week early. She memorises every answer, and scores 100 out of 100 in the exam. Has she learned anything?

A model tested on its own training data does exactly this. The score it reports is a measure of memory, not of ability.

6 Hiding data from your own model

Train and test split

Before training, put some rows aside. The model never sees them. You score it only on those hidden rows.

Practice papers to study from are the **training set**. The real, unseen exam is the **test set**. Typically eighty percent train and twenty percent test.

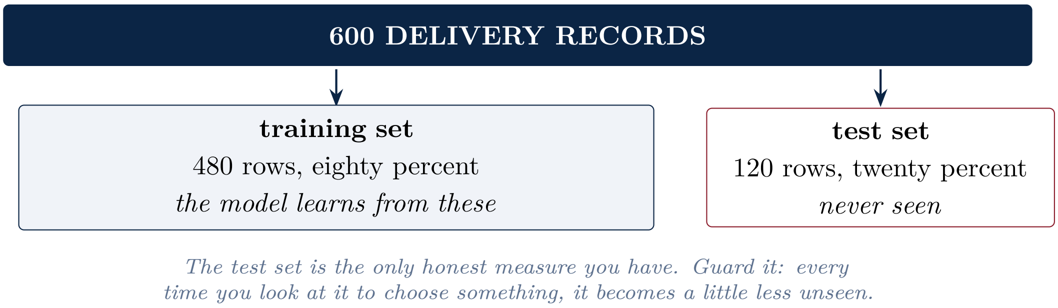


Figure 3: Our 600 rows, split. The proportions matter less than the discipline: the model must be scored on rows it has never met.

6.1 Splitting, in three lines

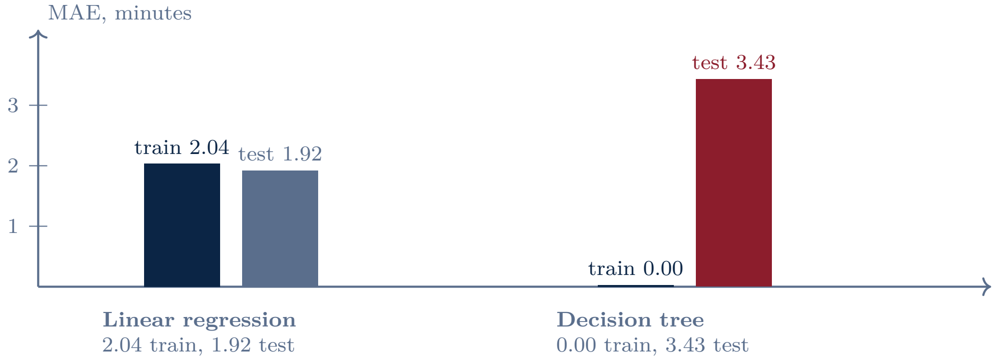
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

`test_size=0.2` keeps one row in five for the exam. `random_state=42` fixes the shuffle, so that you and your teammate get the same split. Without it, every run gives a different score, and nobody can compare two models honestly, because they were never scored on the same rows.

7 What overfitting actually looks like

Two models on our own data, scored both ways with mean absolute error, in minutes.

Model	MAE on train	MAE on test
Linear regression	2.04	1.92
Decision tree, no depth limit	0.00	3.43



The tree is perfect on data it has seen. It memorised it. On unseen orders it is nearly twice as bad as the simple model, while the linear model's two numbers agree with each other.

Figure 4: Overfitting, measured on our own dataset. The healthy sign is not a low training score; it is two numbers that agree with each other.

The reading, in one line

A perfect training score is not an achievement. It is a symptom.

7.1 Why it happens, and what the curve looks like

As a model is allowed to become more complex, its training error keeps falling, because it can bend to fit every row it has seen. Its test error falls at first, because it is learning real structure, and then rises, because it starts learning the noise particular to those rows.

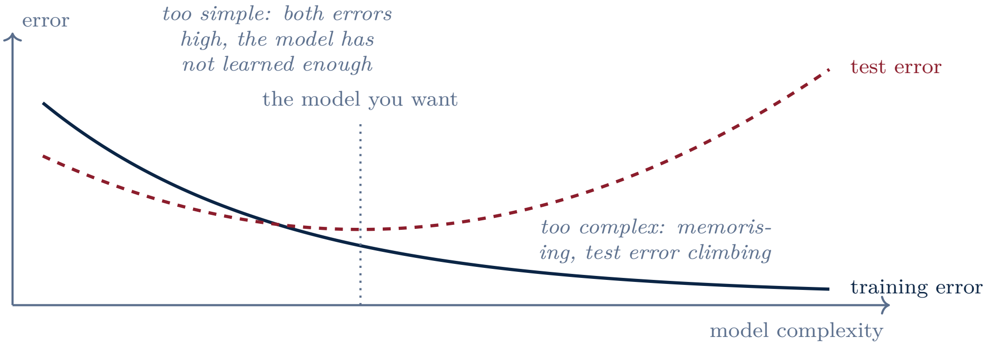


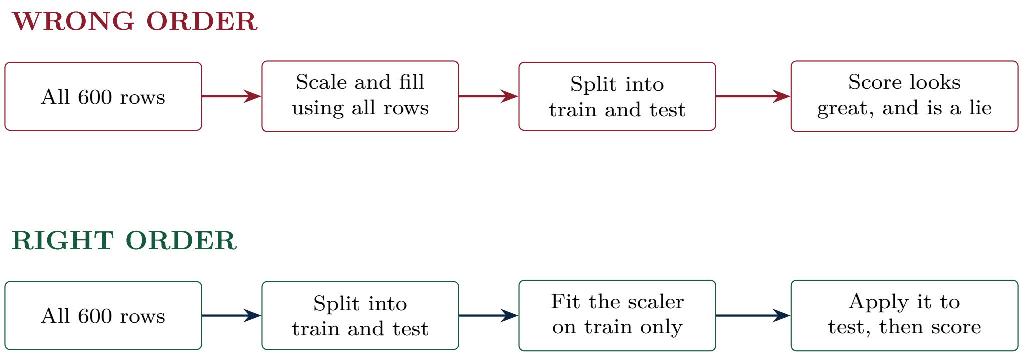
Figure 5: The shape every practitioner carries in their head. You cannot see the dashed curve without a test set, which is the entire argument for hiding data from yourself.

8 Leakage: the error that flatters you

Definition

Data leakage is the use of information when building the model that would not be available at prediction time. It produces scores that are too good, and a system that disappoints in production.

The most common form in a student project is ordering: scaling, or filling missing values, computed over the whole dataset *before* splitting. The test rows then influenced the training, and the test set is no longer unseen.



The scikit-learn rule, stated in its own documentation: never call fit on the test data. A Pipeline enforces the right order for you, which is why every later practical in this course uses one.

Figure 6: The same four operations in two orders. Only the sequence differs, and only one of the two produces a number you can put in a report.

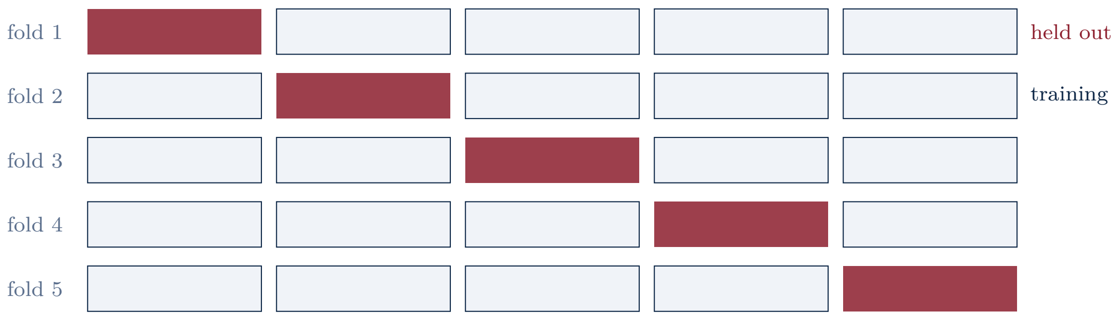
This is not a student problem. It is a research crisis

Kapoor and Narayanan surveyed the scientific literature that uses machine learning and found leakage in seventeen fields, collectively affecting at least 294 papers, in some cases producing wildly overoptimistic conclusions. They classify the failures into eight distinct types and propose model info sheets as a preventive measure. The paper is open access.

Source: S. Kapoor and A. Narayanan, *Leakage and the reproducibility crisis in machine learning based science*, Patterns 4(9), 100804, 2023. `https://doi.org/10.1016/j.patter.2023.100804`

9 Beyond a single split: cross validation

A single split gives you one number, measured on 120 rows. Change the shuffle and the number moves. Cross validation reduces that luck: the data is divided into k folds, and the model is trained k times, each time holding out a different fold for scoring. You report the mean and the spread.



Five folds, five models, five scores. Report the mean and the spread rather than a single lucky number. The cost is five times the training time, which on our dataset is a few seconds.

Figure 7: Five fold cross validation. Every row is used for training four times and for scoring exactly once, which is why the estimate is steadier than a single split.

10 Common errors in splitting

The error	What actually happened
Scoring on the training set and reporting that number	You reported the memorised score, not the real one
Splitting after scaling or filling missing values	The test rows influenced the training. This is leakage
Forgetting `random_state`, then arguing about which model is better	Only the shuffle changed. You compared splits, not models
Peeking at the test set repeatedly to pick a model	It stops being unseen. Each look spends a little of its value
Using a random split on data with a time order	Tomorrow's rows train a model scored on yesterday's. In production the arrow only points one way

PART III

Scoring Honestly

11 Accuracy is not one number

For a regression model, how good is it has several answers, and they are not interchangeable.

- How wrong is it *typically*?
- How wrong is it at its *worst*?
- Is it better than doing nothing at all?

Different scores answer different questions. Pick one before you build, and keep it.

12 Two scores, both in minutes

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\, y_i - \hat{y}_i \,| \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

MAE is the average size of the mistake. RMSE is the same idea with large mistakes punished much harder, because the errors are squared before they are averaged. Both come out in minutes, so both are easy to explain to somebody who does not build models.

Two facts worth memorising

RMSE is always at least as large as MAE. And a **large gap between them means a few very bad predictions**, because only the squaring can produce that gap.

SMALL GAP

MAE 1.92, RMSE 2.48

errors of a similar size throughout

A steady model. What you see on the test set is roughly what each customer will experience.

LARGE GAP

MAE 2.00, RMSE 4.90

mostly excellent, occasionally disastrous

Go and look at the worst predictions. There is usually a pattern, and it is usually a slice you forgot.

Figure 8: The gap between the two scores is itself a diagnostic. Reporting only one of them throws that diagnostic away.

12.1 Measuring both

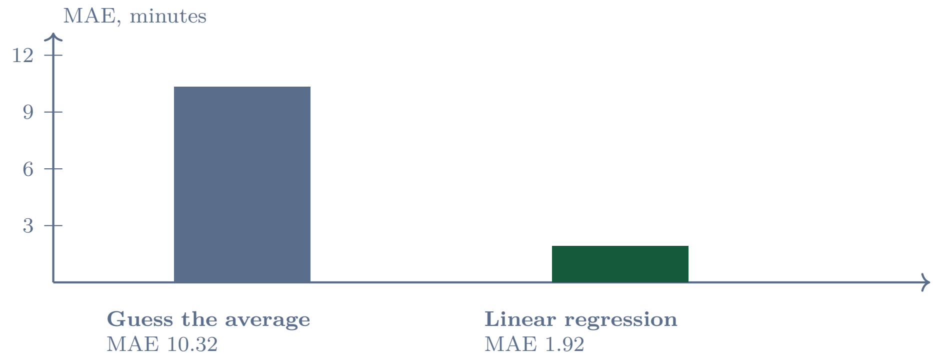
```
from sklearn.metrics import mean_absolute_error, mean_squared_error
import numpy as np

preds = model.predict(X_test)
mae  = mean_absolute_error(y_test, preds)
rmse = np.sqrt(mean_squared_error(y_test, preds))
```

Always on `X_test`, the rows the model never saw. The squared error function returns the squared value, so take the square root yourself to get back to minutes.

13 The number to beat

Before celebrating a model, ask what doing nothing would score. On our 120 test orders:

On the 120 test orders	MAE	RMSE
Always guess the average time	10.32	12.39
Linear regression	1.92	2.48



The model is wrong by about two minutes; guessing is wrong by ten. That comparison, not the two on its own, is the sentence to put in your report.

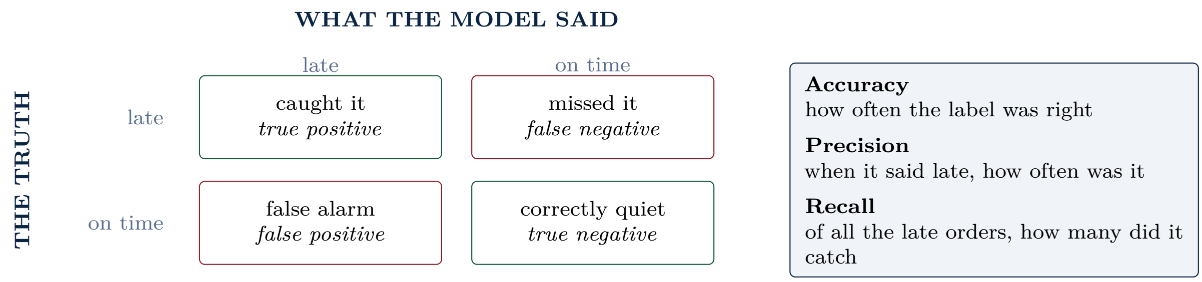
Figure 9: The baseline is what converts a score into evidence. An MAE of 1.92 means nothing until the reader knows that doing nothing scores 10.32 on the same 120 orders.

The sentence to write

On 120 orders the model had never seen, it was wrong by about two minutes on average, against ten minutes for always guessing the mean. Anybody, technical or not, can read that and know what it means.

14 If it had been classification

Lecture 04 uses these, so meet them once. Suppose we had asked the other question: will this order be late?



Precision and recall read the same table in two directions. Which one matters is a business question: a false alarm annoys a customer, a miss breaks a promise.

Figure 10: The confusion matrix, and the three scores read from it. Regression has no equivalent, which is why accuracy cannot be quoted for our project.

Why accuracy can lie

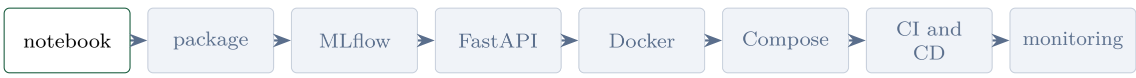
If only 5 orders in 100 are late, a model that always says on time is 95 percent accurate, and completely useless. It never once identifies a late order, which was the only reason anybody wanted the model.

This is why accuracy on an imbalanced problem must never be quoted without precision and recall beside it.

15 Common errors in scoring

The error	The correction
No baseline	An MAE of 4 minutes means nothing until you know that guessing scores 10
Changing the score halfway because the new one looks better	Agree it once, at the start, and record it in the project brief
Comparing two models on different splits	Then you compared the splits, not the models
Quoting accuracy on an imbalanced problem without precision and recall	The majority class is doing the work, not the model
Reporting to three decimal places	An MAE of 1.923 claims a precision the 120 row test set cannot support

16 What our project gains today



The project now has its first working model and an honest number: wrong by about 1.9 minutes on orders it has never seen, against 10.3 for guessing. Every later lecture must keep that number trustworthy.

Figure 11: Still a notebook, but now a notebook with evidence in it. Nothing later in the course is allowed to break the honesty of that number.

PART IV

Consolidation

17 Assignment questions, as released

Set 1 • Regression or classification • CO2

Q1 (Understand). Differentiate regression and classification in your own words. Give two examples of each from an application you use, and say what the model's output looks like in each case.
Q2 (Apply). For each of these, say regression or classification and give a one line reason: (a) predicting tomorrow's temperature, (b) marking an email as spam, (c) predicting how many orders a restaurant gets on Sunday, (d) deciding whether a delivery will be late.
Q3 (Build). Take the delivery time dataset and write down two different questions you could ask of it, one regression and one classification. For each, state the exact column you would predict, and explain in two lines who would use that answer and how.

Set 2 • Train and test • CO2

Q1 (Understand). Explain why a model must be scored on data it has not seen. Then explain what overfitting is, using the exam analogy from this lecture.
Q2 (Apply). A classmate reports 99 percent accuracy and is delighted. You discover the score came from the training rows. Explain in three sentences what is wrong, and write the exact change to the code that would give an honest number.
Q3 (Build). Split the delivery time data eighty and twenty with `random_state=42`. Train any model twice, once scoring on the training set and once on the test set, and report both numbers in a small table. Write two lines explaining which one you would put in a report, and why.

Set 3 • Scoring honestly • CO2

Q1 (Understand). Explain the difference between MAE and RMSE, and say what it tells you when RMSE is much larger than MAE. Then explain why a baseline is needed before either number means anything.
Q2 (Apply). A model scores MAE 4.0 minutes. The always guess the average baseline scores MAE 4.3 minutes on the same test set. Write three sentences on whether this model is worth deploying, and state one extra piece of information you would ask for.
Q3 (Build). On the delivery time test set, compute MAE and RMSE for a model of your choice *and* for the always guess the average baseline. Put all four numbers in a table and write two sentences interpreting them for a non technical reader.

18 Check your understanding

1. State the two kinds of supervised question, and say which one our project is and why.
2. Name one legitimate reason to convert a regression problem into a classification problem.
3. Why is reporting accuracy for the delivery time model a category error rather than a small mistake?

4. Explain the exam analogy for a train and test split in two sentences.
5. Our tree scored 0.00 on train and 3.43 on test. What single word describes this, and which of the two numbers would you put in a report?
6. Sketch the complexity against error curve from memory and label both curves.
7. Give the correct order of operations for splitting and scaling, and say what goes wrong in the other order.
8. What does `random_state=42` buy you, and what argument does it prevent?
9. In five fold cross validation, how many times is each row used for training, and how many times for scoring?
10. MAE is 2.0 and RMSE is 4.9. What does that gap tell you, and what would you do next?
11. Why does an MAE of 1.92 mean nothing on its own?
12. A model is 95 percent accurate at spotting late orders, and 5 percent of orders are late. What might be happening, and which two scores would you ask for?

19 Vocabulary introduced today

Term	Meaning
Regression	Predicting a number on a scale
Classification	Predicting which group something belongs to
Training set	The rows the model learns from
Test set	The rows hidden from the model, used only for scoring
Overfitting	Learning the particular rows rather than the pattern: excellent on train, poor on test
Data leakage	Using information when building the model that would not be available at prediction time
Pipeline	An object that binds preprocessing and model together so that the fitting order cannot go wrong
Cross validation	Repeated splitting into k folds, so the score is a mean with a spread rather than a single lucky number
Baseline	A trivial rule whose score any model must beat
MAE	Mean absolute error, the average size of the mistake, in the units of the target
RMSE	Root mean squared error, the same idea with large mistakes punished harder
Accuracy, precision, recall	Classification scores read from the confusion matrix. None of them applies to a regression model
Confusion matrix	The two by two table of what was true against what was predicted

PART V

The Week Ahead

20 Seven days, about seven hours

Day 1 • The map, then the branch • 45 minutes

Read. scikit-learn, Choosing the right estimator, then the supervised learning chapter of the user guide. https://scikit-learn.org/stable/machine_learning_map.html and https://scikit-learn.org/stable/user_guide.html

Produce. Trace the path from START to the regression branch, and name three estimators you find there that you have never used.

Day 2 • Split it properly • 45 minutes

Read. The `train_test_split` reference page. https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html

Produce. Run the split with three different values of `random_state` and record how much the test MAE moves. That spread is the uncertainty in your single number, and most students have never measured it.

Day 3 • The pitfalls page • 60 minutes

Read. scikit-learn, Common pitfalls and recommended practices. It covers inconsistent preprocessing, data leakage and controlling randomness, which is exactly today's lecture. https://scikit-learn.org/stable/common_pitfalls.html

Produce. One sentence in your own words on why the rule is never call fit on the test data.

Day 4 • Leakage, as a research problem • 75 minutes

Read. Kapoor and Narayanan, *Leakage and the reproducibility crisis in machine learning based science*, Patterns 2023, open access. Read the taxonomy of eight leakage types. https://doi.org/10.1016/j.patter.2023.100804

Produce. Check your own notebook against all eight types and write down which ones could apply to the delivery project.

Day 5 • Metrics, in depth • 60 minutes

Read. The metrics and scoring chapter of the scikit-learn user guide, the regression section. https://scikit-learn.org/stable/modules/model_evaluation.html

Produce. A short table of four regression metrics, each with one line on what it punishes and one line on when you would choose it.

Day 6 • Cross validation • 60 minutes

Read. The cross validation chapter of the user guide, then run `cross_val_score` on the delivery data with five folds.

Produce. Compare the five fold mean against your single split number. If they differ by more than the spread of the folds, say what you think that means.

Day 7 • Write the results table • 60 minutes

Do. Assignment Set 3, Q3: MAE and RMSE for your model and for the baseline, four numbers in one table, and two sentences interpreting them for a non technical reader.

Why this matters beyond the marks. This table is the first artefact in your project that somebody else could audit. From Lecture 11 onward, a tracking server will keep a version of it for every run you ever make.

If you only have one hour all week

Day 3, then Day 7. The pitfalls page prevents the mistake that invalidates everything else, and the results table is the deliverable the next four lectures build on.

PART VI

Resource Library

All links verified on 21 August 2026.

21 For this lecture specifically

Resource	Where, and why
scikit-learn, Choosing the right estimator	https://scikit-learn.org/stable/machine_learning_map.html The decision map used in Section 2
scikit-learn, `train_test_split`	https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html
scikit-learn, Metrics and scoring	https://scikit-learn.org/stable/modules/model_evaluation.html Every regression and classification score, defined precisely
scikit-learn, Common pitfalls and recommended practices	https://scikit-learn.org/stable/common_pitfalls.html Inconsistent preprocessing, data leakage, controlling randomness
scikit-learn user guide	https://scikit-learn.org/stable/user_guide.html Supervised learning, model selection and cross validation chapters
Kapoor and Narayanan, Patterns 2023, open access	https://doi.org/10.1016/j.patter.2023.100804 Leakage across 17 fields and 294 papers, with a taxonomy of eight types
Hyndman and Athanasopoulos, *Forecasting: Principles and Practice*	https://otexts.com/fpp3/ The clearest free treatment of error measures
Google, *Rules of Machine Learning*	https://developers.google.com/machine-learning/guides/rules-of-ml Rule 4 in particular: keep the first model simple and get the infrastructure right

22 Carried forward

Resource	Where
Sculley et al., *Hidden Technical Debt in Machine Learning Systems*, NeurIPS 2015	`https://proceedings.neurips.cc/paper_files/paper/2015/file/86df7dcfd896fcaf2674f757a2463eba-Paper.pdf`
Sato, Wider and Windheuser, *Continuous Delivery for Machine Learning*	`https://martinfowler.com/articles/cd4ml.html`
Amershi et al., *Software Engineering for Machine Learning*, ICSE 2019	`https://www.microsoft.com/en-us/research/wp-content/uploads/2019/03/amershi-icse-2019_Software_Engineering_for_Machine_Learning.pdf`
Paleyes, Urma and Lawrence, *Challenges in Deploying Machine Learning*	`https://arxiv.org/abs/2011.09926`
Machine Learning in Production, DeepLearning.AI, free to audit	`https://www.coursera.org/learn/introduction-to-machine-learning-in-production`
MLOps Zoomcamp, DataTalks.Club	`https://github.com/DataTalksClub/mlops-zoomcamp`
Made With ML	`https://madewithml.com`
Stanford CS 329S	`https://stanford-cs329s.github.io`
Cookiecutter Data Science	`https://cookiecutter-data-science.drivendata.org/`

If you never hid any data, you do not have a score.

Next lecture. Lecture 04, machine learning refresher II: one model is not enough. Logistic regression, k nearest neighbours and support vector machines; random forests, the model our project will actually use; and k means, for when nobody labelled the data. Bring the split and the two scores from today's assignment.